Rituraj Singh

AI PRODUCT ENGINEER · LLM SYSTEMS BUILDER · FULL-STACK SAAS · EX-TATA STEEL

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SUMMARY

AI product engineer with a track record of shipping production systems end-to-end — from schema design to deployed UI. Built Praxya, an AI-powered GHG compliance platform for Indian chemical manufacturers, handling the full pipeline from data ingestion to auditor-ready BRSR reports. Solo-built FileMyCa, a B2B SaaS for Indian CA firms, in 5 days. Built a Conversion Funnel Analyzer using a 3-stage Gemini LLM pipeline with SSRF-safe scraping, deterministic DOM extraction, client-side Fix Simulator and Persona Lens. Deployed computer vision safety systems at Tata Steel. Specializes in agentic pipelines, structured LLM outputs, and multi-tenant SaaS architecture.

TECHNICAL SKILLS

LANGUAGES
Python · TypeScript · JavaScript · Java · C++

LLM / AI
Claude API (tool_use / structured outputs) · Gemini · Groq · LangChain · OpenAI SDK · Pydantic v2 · Agentic Pipelines · RAG

ML / CV
PyTorch · TensorFlow · Scikit-learn · OpenCV · Pandas · Time-series Analysis

FULL-STACK
Next.js 15 · React 19 · FastAPI · Supabase · PostgreSQL · MongoDB · SheetJS · Recharts

INFRA / DEVOPS
Docker · AWS · Vercel · Railway · pg_notify Queue · Supabase RLS · Git · GitHub Actions

PROJECTS

Praxya — AI-Powered GHG Compliance Platform github.com/Rituraj6047/praxya
Mar 2026 - Present

`Python` `Supabase` `Claude API (tool_use)` `Next.js 15` `FastAPI` `Groq · Llama 3.3 70B` `pytesseract` `WeasyPrint` `pg_notify`

- Architected full 9-stage document processing pipeline: PDF upload → OCR (pdfplumber / pytesseract) → LLM extraction via structured tool_use → Pydantic v2 validation → GHG calculation → BRSR-compliant PDF report
- Built stoichiometric GHG calculator (Scope 1 & 2) using IPCC 2006 emission factors + CEA India grid intensity; Decimal-only arithmetic throughout — zero floating-point error in any financial or regulatory output
- Implemented LLM extraction layer using Claude API tool_use mode (migrated to Groq / Llama 3.3 70B for cost); LLM output isolated from calculations via Pydantic schema — zero raw text enters financial data
- Designed multi-tenant Supabase schema with INSERT-ONLY audit trail, RLS across all 8 data tables, pg_notify + polling worker with FOR UPDATE SKIP LOCKED for race-safe concurrent job dispatch

FileMyCa — GST Data Collection SaaS github.com/Rituraj6047/FilemyCA
Feb 2026 - Mar 2026

`Next.js 15` `TypeScript` `Supabase` `SheetJS` `Recharts` `Vercel`

- Solo-built production-ready B2B SaaS portal replacing WhatsApp-based GST data collection for Indian CA firms — shipped in 5 days from scaffold to Vercel deploy
- CA admin dashboard with live status tracking across 47 MSME clients, submission charts, deadline countdown, filterable/sortable client table, and slide-out detail panels with status controls
- MSME client upload portal with SheetJS Excel validation (6 required GST column checks), drag-drop Supabase Storage upload, and 4-state upload machine with success animation
- Reminder system with WhatsApp/email template preview modal, bulk send, per-client 60s anti-spam cooldown, reminder history log, and API stubs architected for Gupshup integration

Conversion Funnel Analyzer — LLM CRO Pipeline github.com/Rituraj6047/conversion-funnel-analyzer
May 2026

`Next.js 15` `TypeScript` `Gemini API` `Cheerio` `Zod` `SSE Streaming` `jsdom / DOMPurify` `Tailwind CSS`

- Built 3-stage sequential LLM pipeline (FunnelMapper → FrictionAnalyzer → Recommender) using Gemini structured output + Zod validation with surgical retry correction on schema failures
- Deterministic Cheerio extractor produces typed PageStructure before any LLM call; cross-validation layer strips hallucinated element IDs and enforces funnel monotonicity client-side
- SSRF-hardened safeFetch with DNS resolution against full private IP blocklist, redirect re-validation on every hop, and 5 MB streaming cap addressing CVE-2024-29415 pattern
- Client-side Fix Simulator and Persona Lens re-run scoring locally with zero additional API calls; results stream via SSE with skeleton states and staggered bar animations

WORK EXPERIENCE

AI Intern · Tata Steel Utilities & Infrastructure Services Ltd
May - Jun 2025

Jamshedpur, India

- Built OpenCV + Python computer vision system detecting speeding vehicles from 4 simultaneous CCTV feeds with real-time frame-by-frame analysis and automated alert generation
- Deployed solution processing 1,000+ vehicle detections per day at 92% accuracy, reducing manual safety monitoring workload by ~60% across an active industrial facility

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) · VIT Bhopal University
Expected May 2027

CGPA: 8.46 / 10.0 · Coursework: Machine Learning · Deep Learning · Computer Vision · NLP · Cloud Computing